Safe Neural Control for a Stationary Distribution of Stochastic Functional Differential Equations

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Outline

1. Problem and motivation
2. Main results
3. Numerical experiments
   - Learning and verification
   - Example 1: Linear system
   - Example 2: Cubic system
   - Example 3: Duffing oscillator
4. Summary

Delayed Feedback & Stability in Distribution

- **Delay feedback for hybrid SDEs**
  You, Hu, Lu & Mao, IEEE Trans. Autom. Control, 2022.
- **Delay feedback for hybrid SDDEs**
  Li, You, Lu, Jiang, Hu & Mao, Int. J. Syst. Sci., 2023.
- **Highly nonlinear hybrid SDDEs**
  Gao, Li & Fan, IEEE Trans. Autom. Control, 2024.
- ...

Control Barrier Functions

- **Safety and performance for ODEs**
  Ames, Grizzle & Tabuada, CDC, 2014;
  Ames, Xu, Grizzle & Tabuada, IEEE TAC, 2017; $\cdots$
- **CBFs for stochastic systems**
  Clark, Automatica, 2021;
  Santoyo, Dutreix & Coogan, Automatica, 2021;
  Nishimura & Hoshino, IEEE TAC, 2024; $\cdots$
- **Safety and stabilization for SDEs and SDDEs**
  Pan, Ren & Sun, IEEE TAC, 2024;
  Zhang et al., ICLR, 2023 (SYNC);
  Zhang et al., ICML, 2024 (FESSNC); $\cdots$

Control Objectives

Design a history-dependent feedback control u to achieve:

- **Stability in distribution:** A unique invariant distribution that attracts the laws of the segment process with an exponential rate.
- **Safety forward invariance:** Almost sure forward invariance of the prescribed safe set $\mathcal{S}$, i.e., trajectories remain safe for all time.

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Controlled SFDE Dynamics

Basic setting and notation

- Let W be a d-dimensional Brownian motion on a filtered probability space satisfying the usual conditions.
- For a delay $\tau > 0$, the history space is $\mathcal{C} = C([-\tau, 0]; \mathbb{R}^d)$ with the supremum norm $\|\phi\| = \sup_{s\in[-\tau,0]} |\phi(s)|$.
- The segment $X_t \in \mathcal{C}$ records the recent history: $X_t(s) = X(t+s)$, $s \in [-\tau, 0]$.
- f is the original drift, u the feedback control, and g the diffusion coefficient.

Controlled SFDE

$$\mathrm{d}X(t) = \bar{f}(X_t)\mathrm{d}t + g(X_t)\mathrm{d}W(t).$$

where $\bar{f} := f + u$ denotes the controlled drift.

Stability in Distribution for SFDEs

For $V, U \in C^2(\mathbb{R}^d; \mathbb{R}_+)$ and $\phi, \psi \in \mathcal{C}$, define the operators:

$$\mathcal{L}V(\phi) = \langle V_x(\phi(0)), \bar{f}(\phi)\rangle + \tfrac{1}{2}\operatorname{tr}[g(\phi)^\top V_{xx}(\phi(0))g(\phi)],$$

$$\mathfrak{L}U(\phi,\psi) = \langle U_x(\phi(0)-\psi(0)), \Delta\bar{f}\rangle + \tfrac{1}{2}\operatorname{tr}[(\Delta g)^\top U_{xx}(\phi(0)-\psi(0))\Delta g].$$

Under local monotonicity and boundedness (Assumptions 1–2):

Assumption 3 (Dissipativity): $\lambda_1 > \lambda_2 \geq 0$,

$$\mathcal{L}V(\phi) \leq L_2 - \lambda_1 \mathcal{W}_1(\phi(0)) + \lambda_2 \int_{-\tau}^{0} \mathcal{W}_1(\phi(s))\mu_1(\mathrm{d}s).$$

Assumption 4 (Contraction): $\lambda_3(x,y) > \lambda_4 \geq 0$ $(x \neq y)$,

$$\mathfrak{L}U(\phi,\psi) \leq -\lambda_3(\phi(0),\psi(0))\mathcal{W}_2(\phi(0),\psi(0)) + \lambda_4 \int_{-\tau}^{0} \mathcal{W}_2(\phi(s),\psi(s))\mu_2(\mathrm{d}s).$$

Under Assumptions 1–4, the system admits a unique invariant distribution and is stable in distribution.[1]

[1] Wang, Wu & Mao. *Syst. Control Lett.*, 2019.

Safety Conditions

Let $H \in C^2(\mathbb{R}^d;\mathbb{R})$ define safe set $\mathcal{S} = \{x : H(x) \geq 0\}$ and its interior $\mathrm{Int}(\mathcal{S}) = \{x : H(x) > 0\}$.

Let $\lambda : (-a, \infty) \to \mathbb{R}$ be continuous and strictly increasing, with $\lambda(0) = 0$ and $\lambda \in C^1(0,\infty)$. Impose the Osgood condition and define Ψ by

$$\Psi(s) = \int_s^1 \frac{\mathrm{d}r}{\lambda(r)} \longrightarrow \infty \quad (s \to 0).$$

Assumption 5: For all $\phi \in C([-\tau, 0]; \mathrm{Int}(\mathcal{S}))$,

$$\mathcal{L}(\Psi \circ H)(\phi) \leq 1 + c_1 \Psi(H(\phi(0))) + c_2 \int_{-\tau}^{0} \Psi(H(\phi(s)))\mu(\mathrm{d}s),$$

where $c_1, c_2 \geq 0$ and μ is a probability measure on $[-\tau, 0]$.

Almost Sure Forward Invariance

Theorem 1

Suppose the controlled SFDE admits a unique global solution and satisfies Assumption 5. For every initial history $\xi \in C([-\tau, 0]; \mathrm{Int}(\mathcal{S}))$,

$$\mathbb{P}\big[X(t) \in \mathcal{S} \text{ for all } t \geq 0\big] = 1.$$

Decoupled Controller Architecture

Boundary singularity dilemma: For $\Psi(s) = -\log s$, Itô's formula yields:

$$\mathcal{L}(-\log H)(\phi) = -\frac{\mathcal{L}H(\phi)}{H(\phi(0))} + \frac{|g^\top(\phi)H_x(\phi(0))|^2}{2H(\phi(0))^2}.$$

Decoupled controller architecture:

$$u_\theta(\phi) = v_\theta(\phi) + w(\phi),$$

where v_θ is smooth and globally Lipschitz.

Normalize $0 < H \leq 1$ on $\mathrm{Int}(\mathcal{S})$. For constants $a, K, \eta > 0$, the barrier term w satisfies:

$$|w(\phi)| \leq \frac{a}{H(\phi(0))}, \qquad \mathcal{L}(-\log H)(\phi) \leq K - \frac{\eta}{H(\phi(0))^2}.$$

Ergodicity and Mixing Time

Theorem 2

Under Assumptions 1–5 on compact $\mathcal{S}$, the segment process has a unique invariant law π on $C([-\tau, 0]; \mathcal{S})$. Define the ε-mixing time by

$$\tau_\varepsilon := \inf\{t \geq 0 : W_1(\mu_t^\xi, \pi) \leq \varepsilon\}.$$

Then

$$W_1(\mu_t^\xi, \pi) \leq C\mathrm{e}^{-\kappa t} \implies \tau_\varepsilon \leq \frac{1}{\kappa} \log\left(\frac{C}{\varepsilon}\right) =: T_\varepsilon.$$

Upper Bound on Control Energy

Theorem 3

Suppose the controlled SFDE satisfies Assumptions 1–5 on compact $\mathcal{S}$, with Assumption 3 holding for $V(x) = \mathcal{W}_1(x) = |x|^2$.

For a deterministic interior initial history ξ and $0 < \varepsilon < 1$, there exist $A_0, B_0 > 0$, independent of ε, such that

$$\mathbb{E}\left[\int_0^{\tau_\varepsilon \wedge T_\varepsilon} |u_\theta(X_s)|^2 \mathrm{d}s\right] \leq \frac{A_0}{\kappa} \log\left(\frac{C}{\varepsilon}\right) + B_0.$$

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Gating Mechanism

In the numerical experiments, we consider the following SDDE with $x = X(t)$ and $y = X(t-\tau)$:

$$\mathrm{d}X(t) = \big[F(x,y) + u_\theta(x,y)\big]\mathrm{d}t + G(x,y)\mathrm{d}W(t).$$

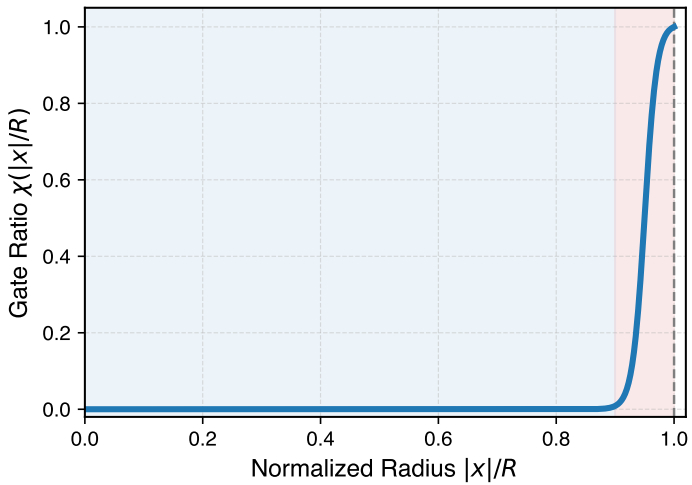
Take $\mathcal{S} = \{|x| \le R\}$, $H(x) = R^2 - |x|^2$ and $\rho = |x|/R$.

Safety term w

$$w(x) = -\frac{6x}{H(x)}\chi(\rho),$$

$$\chi(\rho) = \frac{\sigma\left(5\frac{\rho^2-0.95^2}{1-0.95^2}\right)}{\sigma(5)},$$

$$\sigma(z) = \frac{1}{1+\mathrm{e}^{-z}}.$$



Neural Controller

In $u_\theta(x, y) = v_\theta(x, y) + w(x)$, the analytic safety term $w(x)$ is fixed. We parameterize:

- **Learned feedback law** $v_\theta(x, y)$**:**
  - **Linear systems:** state feedback $v_\theta(x, y) = K_1 x + K_2 y$ with gains $\theta = \{K_1, K_2\}$.
  - **Nonlinear systems:** MLP $\mathcal{N}_\theta(x, y)$ with $\tanh$ activations on concatenated states $\xi = (x, y) \in \mathbb{R}^{2d}$. Smoothness ($\tanh \in C^\infty$) ensures $v_\theta \in C^2$.
- **Neural Lyapunov-like function** $V_\zeta(x, \bar{x})$**:**

  $$V_\zeta(x, \bar{x}) = z^\top \left[\mathcal{N}_\zeta(z)\mathcal{N}_\zeta(z)^\top + \epsilon_V I_d\right] z, \quad z = x - \bar{x}, \quad \epsilon_V > 0.$$

  Structurally guarantees positive definiteness: $V_\zeta(x, \bar{x}) \geq \epsilon_V |z|^2 > 0$ for $z \neq 0$.

Loss Function

Sample (x, y) on $\widetilde{D} = \{x : |x| \le 0.99R\}$. Empirical penalty $\hat{L} = \frac{1}{N}\sum_{i=1}^{N} \max(0, \mathcal{R}^{(i)})^2$:

- **Safety loss $\hat{L}_{\text{safe}}(\theta)$:**

$$\mathcal{R}_{\text{safe}} = -\mathcal{A}H + \frac{|G^\top H_x|^2}{2H} - H\big[1 + c_1 \ln(R/H(x)) + c_2 \ln(R/H(y))\big].$$

- **Dissipativity loss $\hat{L}_{\text{dissip}}(\theta)$:**

$$\mathcal{R}_{\text{dissip}} = 2\langle x, \bar{F}\rangle + |G|^2 - a_1 + a_2|x|^\alpha - a_3|y|^\alpha.$$

- **Contraction loss $\hat{L}_{\text{contract}}(\theta, \zeta)$:**

$$\mathcal{R}_{\text{contract}} = 2\langle \Delta x, \Delta \bar{F}\rangle + |\Delta G|^2 + b_1|\Delta x|^2 - b_2|\Delta y|^2 + b_3 V_\zeta - b_4 \bar{V}_\zeta.$$

Total loss: $L_{\text{total}}(\theta, \zeta) = \lambda_{\text{exist}}(\hat{L}_{\text{dissip}} + \hat{L}_{\text{contract}}) + \lambda_{\text{safe}}\hat{L}_{\text{safe}}$.

[2] SDDE stationarity criteria: Li, Mao & Song. *J. Theor. Probab.*, 2024.

SMT-Guided Counterexample Training

Step 1. Learning

- Train feedback v_θ and Lyapunov-like function V_ζ on sampled states.

Step 2. Falsification[3]

- Freeze weights; use dReal to search for violations of safety, dissipativity, or contraction.

Step 3. Decision

- **If counterexamples found:** add them to the training set and return to **Step 1**.
- **Otherwise (verified):** all conditions hold on the full domain $\Longrightarrow$ accept controller.

[3] Chang, Roohi, Gao. Neural Lyapunov control. NeurIPS, 2019.

Verification Summary

For each frozen controller, the safety, dissipativity and contraction inequalities were verified throughout the open ball.

System	Radius R	Adam updates	Replay pool
Linear	1	200	0
Cubic	1	1200	0
Duffing	2	6300	30

Linear and cubic passed at their first verification. Duffing required repeated checks and accumulated 30 confirmed counterexamples before the final weights passed.

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Unstable Linear SDDE

The uncontrolled SDDE with $\tau = 0.5$ is

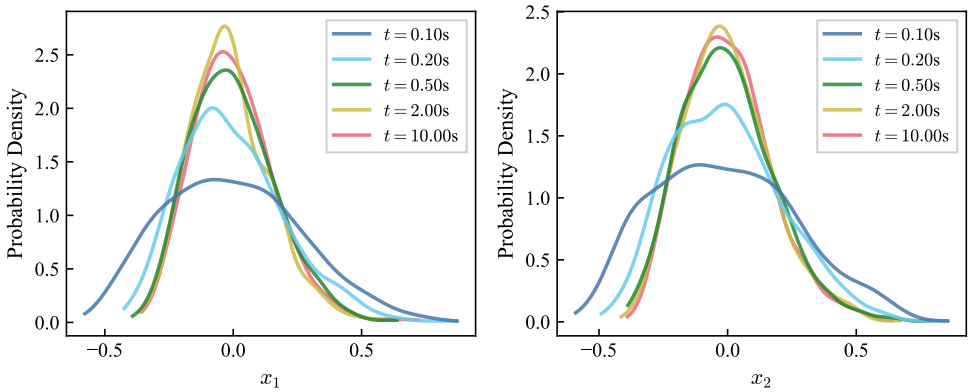
$$\mathrm{d}X(t) = F(x, y)\mathrm{d}t + G(x)\mathrm{d}W(t),$$

where

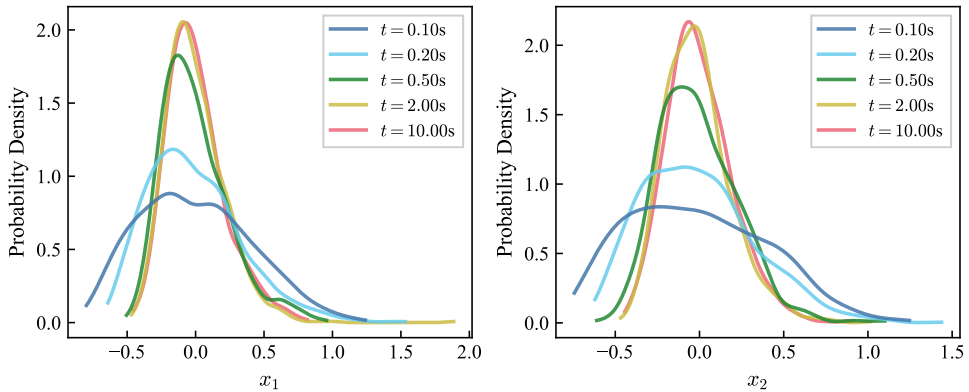
$$F(x, y) = Ax + By, \qquad A = \begin{bmatrix} 0.5 & 1 \\ -1 & 0.2 \end{bmatrix}, \qquad B = \begin{bmatrix} 0.3 & 0 \\ 0.1 & 0.1 \end{bmatrix},$$

$$G(x) = \begin{bmatrix} 0.5 + 0.5x_1 & 0.25x_2 \\ 0.25x_1 & 0.5 + 0.5x_2 \end{bmatrix}.$$

Linear SDDE: Marginal PDF Evolution

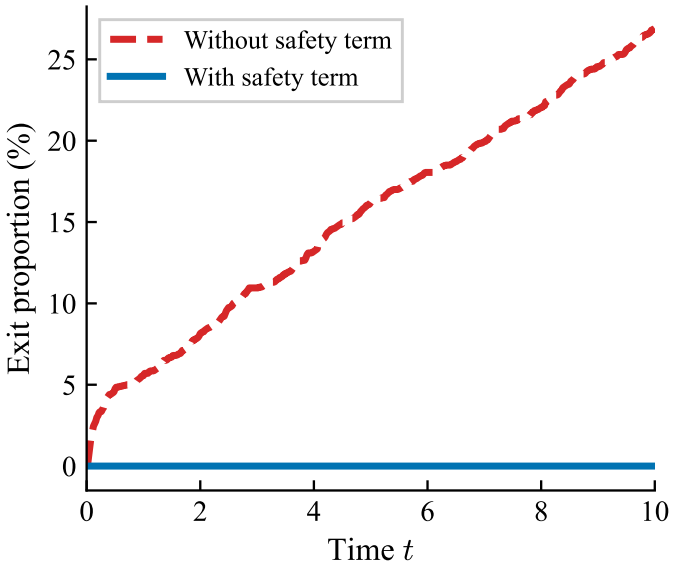


With safety

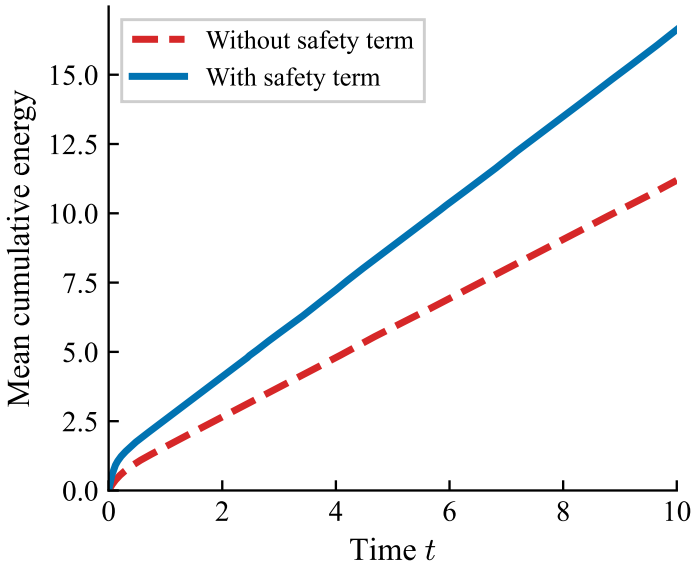


Without safety

Linear SDDE: Safety Comparison



Exit proportion



Cumulative control energy

2000 paths under identical noise and initial histories, $T = 10$:

- **Safety**: 26.9% of the paths exit without the safety term, none with it.
- **Energy**: the safety term raises the mean energy moderately, from 11.2 to 16.6.

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Two-Dimensional Cubic SDDE

The uncontrolled SDDE with $\tau = 0.5$ is

$$\mathrm{d}X(t) = F(x, y)\mathrm{d}t + G(x, y)\mathrm{d}W(t),$$

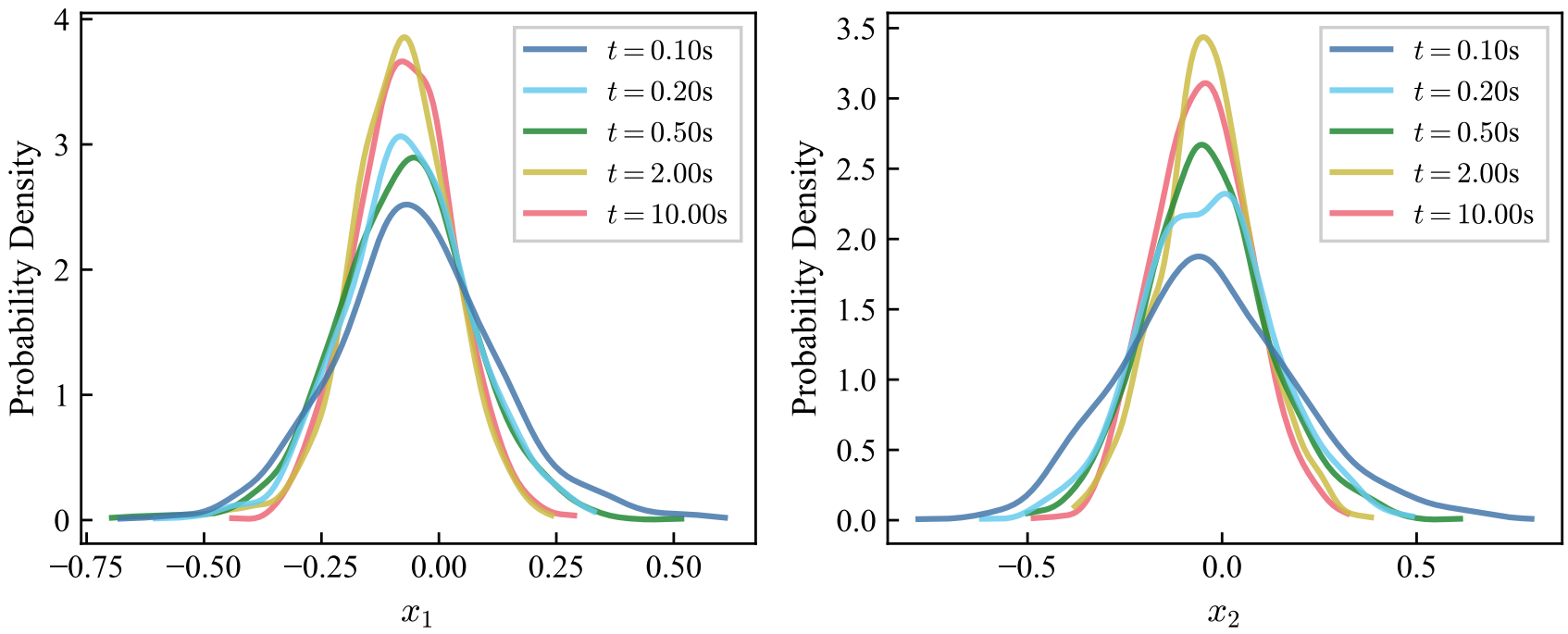
where

$$F(x, y) = \begin{bmatrix} 1.5x_1 + x_1^3 + 0.5x_2 + 0.15y_1 \\ 0.8x_1 + x_2^3 + 1.2x_2 + 0.1y_2 \end{bmatrix},$$

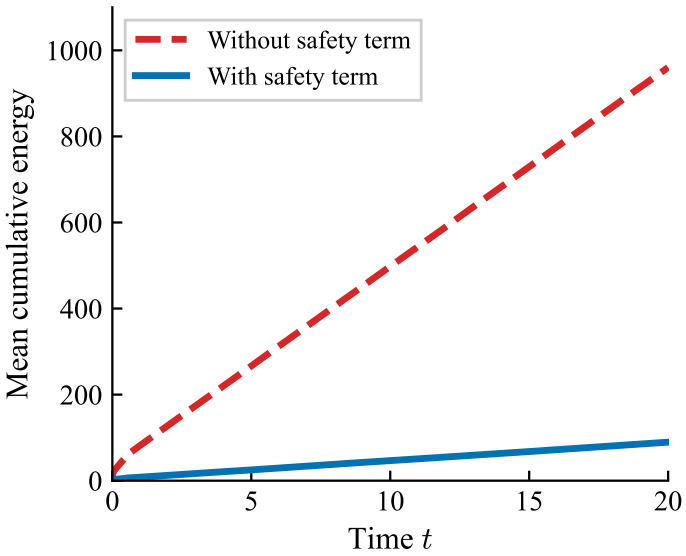
$$G(x, y) = \begin{bmatrix} g_1(x, y) & 0.2x_1x_2 \\ 0.2x_1x_2 & g_2(x, y) \end{bmatrix},$$

$$g_i(x, y) = 0.5 + 0.1|x_i|^{3/2} + 0.5|y_i|, \quad i = 1, 2.$$

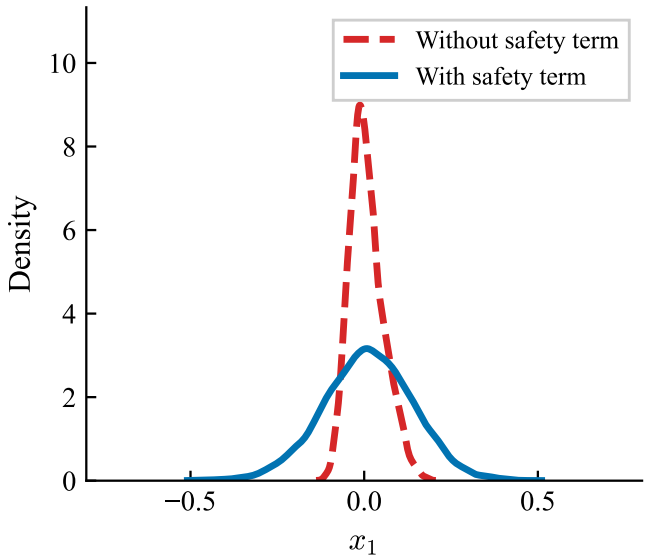
Cubic SDDE: Marginal PDF Evolution



Cubic SDDE: Safety Comparison



Cumulative control energy



Late-time distribution (x_1)

Comparison: without safety term ($w \equiv 0$) vs. with safety term $w(x)$:

- **Energy reduction**: Mean cumulative energy drops by $\approx 90\%$ (from ≈ 961 to 89).
- **Distribution preservation**: Eliminates over-conservatism, restoring the natural spread.

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Duffing Oscillator

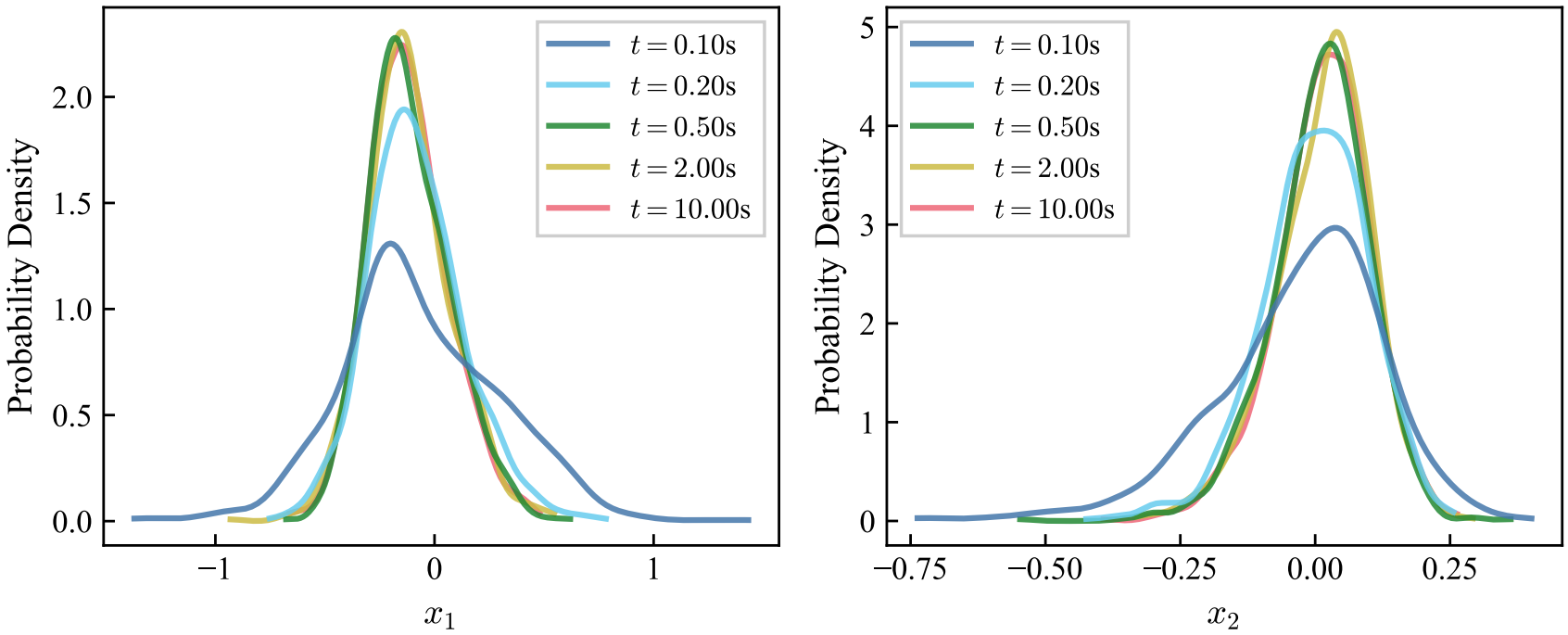
The uncontrolled SDDE with $\tau = 0.5$ is

$$\mathrm{d}X(t) = F(x, y)\mathrm{d}t + G(x, y)\mathrm{d}W(t),$$

where

$$F(x, y) = \begin{bmatrix} x_2 + 0.15y_1 \\ -0.3x_2 - x_1 - x_1^3 + 0.1y_2 \end{bmatrix},$$
$$G(x, y) = \mathrm{diag}\big(1 + 0.2x_1^2,\ 1 + 0.3|x_2|\big).$$

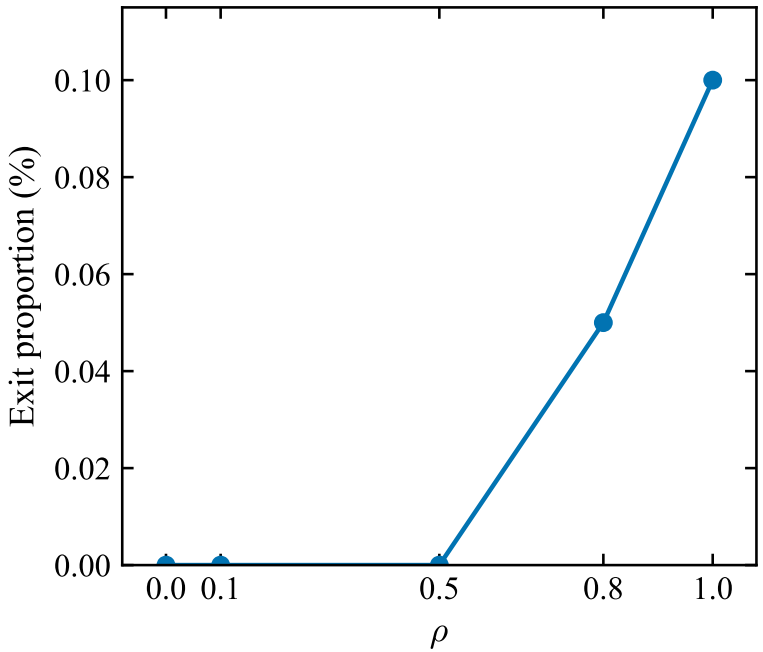
Duffing Oscillator: Marginal PDF Evolution



Duffing Oscillator: Robustness under Perturbations

For perturbation parameter $\rho \in [0, 1]$:

$$F_\rho(x, y) = \begin{bmatrix} x_2 + 0.15(1+\rho)y_1 \\ -0.3(1-\rho)x_2 - x_1 - x_1^3 + 0.1(1+\rho)y_2 \end{bmatrix},$$
$$G_\rho(x) = \operatorname{diag}\big(1 + 0.2(1+\rho)x_1^2,\ 1 + 0.3(1+\rho)|x_2|\big).$$



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https://github.com/zuoruchun/SFNC